AI for Data Visualization

Q1. What is AI in Data Visualization? Explain how it differs from traditional data analysis.

Answer: AI in data visualization uses computer models to find patterns in data and help explain them with charts or written summaries. Traditional analysis usually depends more on people choosing what to compare and interpreting the results themselves. AI can speed up this work, but people still need to check its findings.

Q2. Explain how AI helps in automatic insight generation in BI tools.

Answer: AI in Business Intelligence (BI) tools can scan data and point out changes, unusual values, or relationships. For example, it might highlight that sales fell in one region or that a product sells more in a certain season. This helps users notice patterns faster.

Q3. What are the key AI features available in Microsoft Power BI? Explain any two with examples.

Answer: Power BI includes features such as **Key Influencers**, **Decomposition Tree**, **Smart Narrative**, and **anomaly detection**.

- **Key Influencers** shows which factors are linked with a result. For example, it might show which factors are associated with higher customer satisfaction.
- **Smart Narrative** creates a short written summary of a report. For example, it might describe how this month's sales compare with last month's.

Q4. Explain the concept of Key Influencers and how AI identifies factors affecting a metric.

Answer: Key Influencers helps show which data fields are most connected to a result, such as sales or customer satisfaction. It compares groups in the data to find factors associated with higher or lower values. These are patterns in the data, not proof that one factor caused the result.

Q5. What is the role of Smart Narratives in AI-powered dashboards?

Answer: Smart Narratives turn chart and report results into short written explanations. They can help users understand important changes without examining every chart. Users should still check that the summary matches the data and the business situation.

Q6. Explain how Tableau Pulse provides proactive insights in Tableau.

Answer: Tableau Pulse tracks selected measures, such as sales or profit, and gives users updates about changes and patterns. It can alert users when a measure changes and summarize possible trends, helping them notice important updates sooner.

Q7. What is meant by AI-driven decision-making? How does AI support business decisions?

Answer: AI-driven decision-making means using AI's analysis of data to help people choose what to do. AI can find patterns, make forecasts, and compare possible outcomes. A business can use this information to plan stock, target customers, or manage costs. People should review the results before acting.

Q8. Explain the importance of interpreting AI-generated insights instead of directly trusting them.

Answer: AI can make mistakes when data is incomplete, outdated, or biased. It may also show a relationship that does not explain why something happened. Interpreting an insight means checking the data, considering the real situation, and deciding whether the result makes sense before using it.

Q9. Discuss the limitations of AI in data visualization tools.

Answer: AI tools depend on the quality of the data. Poor or biased data can lead to misleading results. AI may miss important context, confuse correlation with cause, or produce explanations that are hard to understand. It can assist analysis, but people still need to check its output.

Q10. Compare how AI is used differently in Power BI and Tableau.

Answer: Both tools use AI to help users find patterns and understand data.

- **Power BI** includes tools such as Key Influencers, Smart Narratives, and anomaly detection to explain factors, summarize reports, and flag unusual changes.
- **Tableau** includes features such as Tableau Pulse for updates on selected measures, along with tools that help users explore patterns in visualizations.

The best choice depends on the organization’s tools, data, and reporting needs.